Hasan Serdar Arikan

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Computer science master's student in machine learning, artificial intelligence and cybersecurity (3+ years of experience). National Science Foundation SFS scholar. Author of two research publications. U.S. citizen.

Education

Aug 2026 – May 2028

Master of Science in ML, AI and Cybersecurity

University of Nevada, Las Vegas (UNLV)

- National Science Foundation CyberCorps Scholarship for Service (SFS) scholar

May 2021 – Dec 2025

Bachelor of Science in Computer Science

University of Nevada, Las Vegas (UNLV)

- GPA: 3.7

Experience

Jun 2024 – Aug 2024

Machine Learning Intern

Lawrence Berkeley National Laboratory

- Built Python data pipelines to clean and preprocess protein sequence data, delivering accurate, high-quality input for a deep learning model modeled on Google DeepMind's AlphaFold.
- Cut processing time for 90,000+ protein records by 54% through parallelized Python, Bash and Slurm scripting on a high-performance computing cluster.

Jan 2024 – May 2024

Machine Learning Intern

National Renewable Energy Laboratory

- Built and evaluated machine learning models (ARIMA, SVM, ANN, KNN, LSTM) in Python using TensorFlow and Keras to forecast commercial building energy demand.
- Improved forecasting accuracy by 34% over existing baseline models by evaluating alternative algorithms and engineering a feature extraction and data pre-processing pipeline.

Jun 2023 – Aug 2023

High Performance Computing Intern

Lawrence Berkeley National Laboratory

- Designed and executed benchmarking analysis of parallel graph algorithms (triangle counting, connected components, PageRank) on CPU and GPU using C++, OpenMP and Kokkos, with low-level compiler and machine-code optimization.
- Co-authored a peer-reviewed research paper presenting the benchmarking results.

Spring 2023 – Fall 2023

Teaching Assistant

University of Nevada, Las Vegas

- Mentored 60+ students in CS 218 (Systems Programming), guiding them through assembly language, data representation, instruction sets and debugging via weekly office hours and hands-on assignment support.
- Facilitated CS 301 (Social Implications of Computer Technology), leading discussions on computing ethics, data privacy, cybersecurity and intellectual property, and evaluating student presentations and written work.

Projects and involvement

2026 – Present

Cyber Clinic at UNLV

Free Cyber Clinic

- Assessing the cybersecurity posture of local businesses and organizations, identifying vulnerabilities and delivering actionable recommendations to strengthen their security defenses.

Sep 2025 – Dec 2025

Peak Focus: gamified productivity web app and Chrome extension

Team project

- Built a focus-productivity platform in which users climb a pixel-art mountain by completing work sessions, combining a React/TypeScript web app (Vite, Tailwind, Phaser) with a Manifest V3 Chrome extension and a NestJS, Prisma and PostgreSQL backend.
- Architected the Chrome extension (Vite pipeline, background service worker) with session-aware website blocking that persists across restarts, plus Google OAuth 2.0 and a leaderboard spanning app, backend and extension.

Aug 2023 – Dec 2025

Undergraduate Researcher

DiSC Lab, UNLV

- Conducted research at the intersection of large-scale algorithms, high performance computing (HPC) and optimization, developing scalable machine learning models for graph problems.

Publications

Hasan Arikan, et al. (2025). GCN-Driven CUDA Parameter Optimization for Parallel Triangle Counting in Graphs. *IEEE High Performance Extreme Computing Conference (HPEC)*.

Shaikh Arifuzzaman, Hasan Arikan, et al. (2024). Unlocking the Potential: Performance and Portability of Graph Algorithms on Kokkos Framework. *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*.

Skills

Languages

Python, C++, Java, Assembly, SQL, R, Bash

Machine learning

PyTorch, TensorFlow, Keras

HPC and systems

CUDA, OpenMP, Kokkos, Slurm, Linux

Web

React, TypeScript, NestJS, PostgreSQL

Awards and certifications

CyberCorps Scholarship for Service (SFS)

National Science Foundation, 2026

Certified Ethical Hacker (CEH)

Certification

AWS Cloud

Certification (in progress)